

Chapter 21

Waste Management — I Impact of Waste Accumulation

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Any material that is discarded after it has served its purpose and is no longer useful is called

1. compost
2. waste
3. manure
4. pathogen

Answer

waste

Question 2

Any material that is rendered useless during or left over in a manufacturing process is called

1. industrial waste
2. domestic waste
3. commercial waste
4. mining waste

Answer

industrial waste

Question 3

Which of the following is NOT a toxic substances present in e-waste?

1. Cadmium
2. Lead
3. Mercury
4. Methane

Answer

Methane

Question 4

The process of depletion of oxygen from water bodies occurring either naturally or due to human activities is called

1. Biomagnification
2. Oxygenation
3. Eutrophication
4. Acid rain

Answer

Eutrophication

Question 5

The increase in the number of phytoplanktons and algae reduce the penetration of, light and heat into water body.

1. small fish
2. nitrogen
3. oxygen
4. carbon dioxide

Answer

oxygen

Question 6

Domestic waste being organic in nature undergoes and create conditions favourable for the growth of pathogens.

1. fermentation
2. nitrification

3. magnification
4. oxygenation

Answer

fermentation

Question 7

..... is a harmful toxin that affects the development of a child's brain.

1. Lead
2. Mercury
3. Cadmium
4. Zinc

Answer

Lead

Question 8

..... is a chemical that is said to cause cancer.

1. Arsenic
2. Cadmium
3. Calcium
4. Lead

Answer

Arsenic

Question 9

The increase in the concentration of various toxic substances along the food chain is called

1. Eutrophication
2. Bio-accumulation
3. Biomagnification
4. Demagnification

Answer

Biomagnification

Question 10

The phenomenon of concentrated toxic deposition at the higher trophic level is known as

..... .

1. Bio-accumulation
2. Biomagnification
3. Eutrophication
4. Biodegradation

Answer

Bio-accumulation

Question 11

The accumulation of toxins will be higher in which of the following ?

1. Water
2. Algae
3. Small fish
4. Big fish

Answer

Big fish

Question 12

Waste accumulation is a breeding ground for like flies, mosquitoes, rodents and pet animals.

1. hectors
2. vectors
3. spreaders
4. carriers

Answer

vectors

Question 13

Which of the following diseases are Not spread by houseflies?

1. Typhoid
2. Diarrhoea
3. Cholera
4. Amoebiosis.

Answer

Amoebiosis

Question 14

The warming up of the atmosphere due to the increased concentration of Greenhouse Gases is known as

1. Global Warming
2. Climate change
3. Acid rain
4. Radiation

Answer

Global Warming

Question 15

Hepatitis, diarrhoea and corona are diseases.

1. bacterial
2. viral
3. fungal
4. None of the above.

Answer

viral

Answer the following questions

Question 1(a)

What is waste? Name two sources of waste.

Answer

The term 'waste' refers to any material that is discarded because it has served its purpose and is no longer useful.

Two sources of waste are-

1. Domestic waste like food leftovers, bits of paper etc.
2. Industrial waste like paints, sand, fly ash etc.

Question 1(b)

How is the 'use and throw' concept responsible for the increase in waste generation?

Answer

The 'use and throw' concept means to use any product only once and then dispose it.

For example, suppose we purchase a pen. After the ink is finished, we throw it away and buy a new pen instead of buying a refill and reusing the pen. If this example is followed by many people, the amount of waste will definitely increase in volume.

Thus, the 'use and throw' concept is responsible for the increase in waste generation.

Question 1(c)

What is acid rain? State its impact on the environment.

Answer

Acid rain means the presence of excessive acids in rainwater.

Burning of coal, wood and petroleum produce sulphur and nitrogen. These two react with oxygen and are converted into their respective oxides- sulphur dioxide and nitrogen dioxide, which are soluble in water.

During rain, these oxides react with large quantities of water vapour in the atmosphere to form acids like sulphuric acid, sulphurous acid, nitric acid and nitrous acid. These acids, when they precipitate together with rain or snow form acid rain.

Impact of acid rain on the environment are as follows-

1. Acid rain increases the acidity in the soil and destroys forests and crops.
2. It corrodes buildings, monuments, statues, bridges, fences and railings.
3. It poses a serious threat to human health, since it contaminates air and water.
4. Aquatic species are affected due to acid rain.

5. Acid rain affects plant growth. Plant leaves get burnt and dry.

Question 1(d)

What is meant by Ozone layer depletion? How is it harmful?

Answer

Ozone layer depletion is the thinning of the ozone layer present in the upper atmosphere. This happens when the chlorine and bromine atoms in the atmosphere come in contact with ozone and destroy the ozone molecules.

It is harmful because the ozone layer absorbs ultraviolet rays of the sun. But depletion of ozone layer allows the ultraviolet rays of the sun to reach the Earth directly without any obstacle or filtration.

Ultraviolet rays can harm the environment in the following ways-

1. They cause many diseases like skin cancer and cataract.
2. They can cause genetic disorders which ultimately affect heredity.
3. They disturb the ecological balance in marine ecosystem.
4. They can damage physical and chemical properties of any complex chemical substance.

Question 2(a)

What is E-waste? Why is it increasing at a fast rate?

Answer

E-waste or electronic waste refers to various forms of electric and electronic equipment that have ceased to be of any value to its users because they have exhausted their utility value due to redundancy, replacement or breakage.

With advancement in Information Technology, new electronic goods are being produced at a fast rate and this results in an increase in volume of obsolete electronic products. In fact, e-waste is one of the fastest growing waste, constituting more than 5% of all municipal solid waste.

Question 2(b)

Explain briefly how does the industrial waste reach the human beings and affect their health.

Answer

Industries dealing with chemicals, pulp and paper, food processing, etc. produce various kinds of waste material. The smoke coming out from the factories has small particles of dust, carbon,

metals, other solids, liquids and radioactive materials which get mixed in smoke and pollute the air.

The burning of sulphur in coal or heavy oil in thermal power plants releases sulphur dioxide in the air. Sulphur dioxide combines with oxygen and water to form sulphuric acid in the atmosphere which is a health hazard. Besides, it produces acid rain.

In winter, smog, a combination of smoke and fog envelopes many cities creating health hazards and traffic problems. Smog is harmful for health as it causes asthma, bronchitis, shortness of breath, eye and nose irritation, etc. It also reduces visibility and leads to problems in traffic movement.

Industrial waste like heavy metals or synthetic organic compounds reach water bodies either through direct discharge or by leaching from waste dumps. Among the heavy metals, cadmium, lead, mercury, copper, chromium etc., get into water.

The industrial waste mixed with water makes it poisonous and unhygienic. It may also cause various diseases in humans and animals, if consumed without treatment.

Question 2(c)

Name two toxic particulate materials. State the effect of each on human health.

Answer

1. **Lead** — It affects blood system, causes behavioural disorders and can also cause death.
2. **Nickel** — It causes respiratory problems and lung cancer.

Question 2(d)

Give two differences between toxic and non-toxic waste.

Answer

Toxic waste	Non-toxic waste
Toxic waste is non-biodegradable.	Non-toxic waste is biodegradable.
It poses a serious threat to human health and the environment.	It does not pose a serious threat to human health and the environment.
For example, lead, cadmium, mercury etc.	For example, food leftover, fruit and vegetable peels, paper etc.

Question 3(a)

Explain how eutrophication affects aquatic life.

Answer

Eutrophication is the process of depletion of oxygen from water bodies occurring either naturally or due to human activities.

Under normal conditions, algae and phytoplankton use carbon dioxide, inorganic nitrogen and phosphate from the water as food. They serve as food for the zooplankton, which in turn are eaten by fish.

When nutrients become abundant due to waste accumulation, the growth of phytoplankton and algae increases. They reduce the penetration of oxygen, light and heat into the water body. As a result, the aquatic plants are unable to carry out photosynthesis reducing the content of oxygen in the water. This causes death of most of the aquatic organisms, draining water of all its oxygen.

Question 3(b)

What harm is done by dumping of waste near water bodies?

Answer

Waste dumped near a water source percolates through the soil into the water bodies and contaminates the water. This results in the accumulation of toxic substances in the water bodies and further in the food chain through plants and animals that feed on it.

The contaminated water may cause various water borne diseases like typhoid, cholera etc., if the water is consumed without treatment.

Question 3(c)

Give a reason for each of the following:

- (i) Marine animals and turtles are found dead in many coastal areas.
- (ii) Radioactive waste is more harmful than other waste.
- (iii) Uncontrolled burning of waste causes air pollution.

Answer

(i) Marine animals and turtles are found dead in many coastal areas due to plastic debris in the ocean. More than 80% of this plastic comes from land. It washes out from the beaches and streets. It travels through storm drains into streams and rivers. It flies away from landfills into seas. In the ocean, this plastic waste breaks down into smaller pieces, which are easily ingested by marine animals causing blockages in their digestive tracts and eventual death. Marine turtles

are more susceptible to the effects of consuming marine debris because they have downward facing spines which prevent the possibility of regurgitation.

(ii) Radioactive waste is more harmful than other waste because radioactive wastes cause pollution and damage the environment and the high level products of nuclear wastes remain in the environment for several hundred years.

(iii) Uncontrolled burning of waste creates smoke and other air pollutants that release toxic substances into the environment and cause air pollution.

Question 3(d)

What is biomagnification? What can be its effects on humans?

Answer

Biomagnification refers to the increase in the concentration of various toxic substances along the food chain.

Toxic substances at the level of primary producers get concentrated at each trophic level as they move up the food chain. The phenomenon of concentrated toxic deposition at the higher trophic level is known as bio-accumulation.

For example, if there are traces of toxic chemicals in water, then their concentration in algae will be much higher. When fish eat the algae, the concentration of toxins will increase further. Therefore, accumulation of a small amount of toxic chemicals in water can have a serious impact on the fish that live in it. Further, when the fish are consumed by humans and other animals, these toxins can cause various kinds of diseases in them.

Question 4(a)

Name two diseases which occur because of waste accumulation on land.

Answer

Two diseases which occur because of waste accumulation on land are-

1. Malaria
2. Dengue

Question 4(b)

Name two common diseases caused as a result of gaseous pollution.

Answer

Two common diseases caused as a result of gaseous pollution are-

1. Bronchitis
2. Lung cancer

Question 4(c)

Name three water-borne diseases.

Answer

Three water-borne diseases are-

1. Diarrhoea
2. Cholera
3. Typhoid

Question 4(d)

Explain briefly the need for management of waste.

Answer

The problem with waste is that it remains in our environment. We move waste from one place to another but never get rid of it completely. Nature has recycled waste materials for millions of years. However, human population has increased so rapidly during the last century that the environment is now threatened by our activities and the wastes they produce. Much damage is done to the environment by the pollution of air, degradation of soils and contamination of water sources like rivers, lakes etc.

There is a close connection between the waste, pollution and the damage to the environment. Waste causes pollution, which in turn causes the damage. The dangers posed by global pollution due to accumulation of wastes, threaten the survival of the ecosystem and call for the need to manage waste.

Question 5(a)

Why is the handling of solid wastes a major problem?

Answer

The handling of solid waste is a problem because most disposal methods cause harm to the environment. Both open dumps and landfills may contain toxins that seep into the soil and the water bodies and cause soil and water pollution respectively.

The uncontrolled burning of waste creates smoke and other air pollutants that release toxic substances into the environment and cause air pollution. Scavengers and stray animals invade the

open garbage dumps and spread the waste over a large area, thereby, spreading germs and diseases as well as destroying the beauty of the place.

Question 5(b)

Name the gas produced by the decomposition of accumulated waste. Why is this gas harmful?

Answer

Methane gas is gas produced by the decomposition of accumulated waste.

This gas is highly inflammable, and can cause an explosion if not managed properly.

Question 5(c)

Give a reason for each of the following:

- (i) 'Use and throw' concept generates more waste.
- (ii) Run off from fields leads to death in adjacent water bodies.
- (iii) Birds feeding on agricultural waste are at the brink of extinction.

Answer

(i) 'Use and throw' concept generates more waste because this concept means to use any product only once and then dispose it. Therefore, after using a product only once, it is neither re-used nor recycled, but disposed once and for all. And since people are using this concept in modern times, the generation of waste is increasing to new levels.

(ii) Run off from fields leads to death in adjacent water bodies because the fertilisers and pesticides used in the fields have chemicals. When they reach the water bodies, the toxins present in the waste can kill aquatic organisms directly by sufficiently changing the pH of water, covering the water surface and causing a reduction in dissolved oxygen. The aquatic organisms may also become victims of bio-magnification and bio-accumulation.

(iii) Birds feeding on agricultural waste are at the brink of extinction because feeding on agricultural waste affects them adversely. Such birds have thinner and weaker egg shells and increased mortality.

Question 5(d)

What is Global Warming? Name any two Greenhouse Gases?

Answer

The rise in average mean temperature of the earth on account of enhanced concentration of greenhouse gases in the atmosphere is known as global warming.

Two Greenhouse gases are-

1. Carbon dioxide
2. Methane

Thinking Skills

Question 1

According to a World Bank Report, global waste will increase by 70 per cent by 2050 unless urgent action is taken. Suggest five measures to reduce accumulation of waste.

Answer

Five measures to reduce accumulation of waste are:

1. Implement comprehensive waste reduction and recycling programs to encourage waste separation and divert waste from landfills.
2. Adopt extended producer responsibility (EPR) policies to hold manufacturers accountable for product lifecycle and promote sustainable design and disposal practices.
3. Transition to a circular economy model that emphasizes reusing, repurposing, and refurbishing products, reducing waste generation.
4. Conduct education and awareness campaigns to inform the public about responsible consumption, waste segregation and recycling techniques.
5. Promote composting and organic waste management to divert organic waste from landfills and produce nutrient-rich compost for agricultural purposes.

These measures can help reduce waste accumulation, conserve resources and mitigate the environmental impact of waste generation.

Question 2

Waste accumulation is the most visible form of pollution, which creates global environmental challenges. State any three such challenges and their probable remedies.

Answer

Three global environmental challenges and their probable remedies are as follows:

Land and soil pollution —

Both open dumps and landfills may contain toxins that seep into the soil and cause soil pollution. Scavengers and stray animals invade the open garbage dumps and spread the waste over a large area, thereby, spreading germs and diseases as well as destroying the beauty of the place.

Remedy —

1. Implement strict waste management practices, including proper landfill design, lining, and leachate collection systems.
2. Promote waste reduction, recycling, and composting to minimize the amount of waste sent to landfills.

Air pollution —

Improper waste incineration and open burning release harmful pollutants into the air, contributing to air pollution and respiratory health issues. As accumulated waste decomposes, it produces a large quantity of methane gas. This is highly inflammable, and can cause an explosion if not managed properly.

Remedy —

1. Promote waste-to-energy technologies that use proper incineration methods and emissions control systems.
2. Encourage the adoption of cleaner energy sources and waste management strategies that prioritize recycling and composting over incineration.

Water pollution —

Water pollution occurs when people discharge large amount of waste into water bodies, and the natural cleansing process in the water bodies cannot function properly. The process of eutrophication takes place due to introduction of nutrients and chemicals through discharge of domestic sewage, industrial effluents and fertilizers from agricultural fields. This causes death of most of the aquatic organisms, draining water of all its oxygen.

Remedy —

1. Implement effective waste management systems, including proper waste collection, recycling, and awareness campaigns to discourage littering.
2. Encourage the use of reusable and sustainable materials to reduce plastic waste.